Numbers, Groups and Codes

Week 2: Mathematical Induction and Prime Factorization

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Plan for this week

- mathematical induction
- prime numbers and composite numbers
- prime factorization
- Fundamental Theorem of Arithmetic
- Euclid's Lemma
- gcd and lcm from prime factorization

Why induction?

Many statements in mathematics are indexed by

$$n = 1, 2, 3, \ldots$$

To prove such statements systematically, we use **mathematical induction**.
This is one of the most important proof techniques in mathematics.

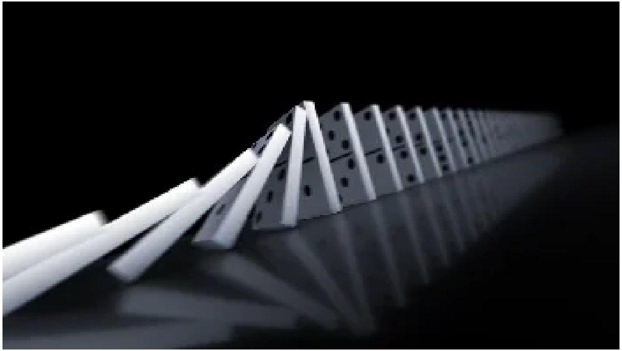
Principle of mathematical induction

Let $P(n)$ be a statement for integers $n \geq 1$.
If

- $P(1)$ is true,
- for every $k \geq 1$, $P(k) \Rightarrow P(k+1)$,

then $P(n)$ is true for all $n \geq 1$.

Domino analogy



- first domino falls
- each domino pushes the next

Structure of an induction proof

An induction proof has two parts:

- **Base case:** prove the statement for the first value
- **Inductive step:** assume it holds for k and prove it for $k + 1$

The assumption that $P(k)$ holds is called the **induction hypothesis**.

A first example

For every integer $n \geq 0$,

$$1 + r + r^2 + \cdots + r^n = \frac{r^{n+1} - 1}{r - 1} \qquad (r \neq 1).$$

We prove this by induction on n.

Geometric series: base case

For $n = 0$,

$$1 = \frac{r-1}{r-1}.$$

So the formula is true for the base case.

Geometric series: inductive step

Assume

$$1 + r + \cdots + r^k = \frac{r^{k+1} - 1}{r - 1}.$$

Then

$$1 + r + \cdots + r^k + r^{k+1} = \frac{r^{k+1} - 1}{r - 1} + r^{k+1}.$$

Geometric series: finish

Continuing,

$$\frac{r^{k+1}-1}{r-1}+r^{k+1}=\frac{r^{k+1}-1+r^{k+1}(r-1)}{r-1}=\frac{r^{k+2}-1}{r-1}.$$

Thus the statement holds for $k+1$.

A number-theoretic example

For every integer $n \geq 1$,

$$6 \mid n(n+1)(n+2).$$

In words: the product of three consecutive integers is always divisible by 6.

Divisibility example: base case

For $n = 1$,

$$1 \cdot 2 \cdot 3 = 6,$$

so the statement is true.

Divisibility example: inductive step

Assume

$$6 \mid k(k+1)(k+2).$$

We want to prove

$$6 \mid (k+1)(k+2)(k+3).$$

Observe that

$$(k+1)(k+2)(k+3) - k(k+1)(k+2) = 3(k+1)(k+2).$$

Divisibility example: finish

Now $(k+1)(k+2)$ is the product of two consecutive integers, so it is even.
Hence

$$3(k+1)(k+2)$$

is divisible by 6.
Therefore

$$(k+1)(k+2)(k+3)$$

is divisible by 6.

Another standard example

For every integer $n \geq 0$,

$$2^n \geq n + 1.$$

This is a simple but useful induction proof involving inequalities.

Inequality example: proof

For $n = 0$,

$$2^0 = 1 \geq 1.$$

Assume

$$2^k \geq k + 1.$$

Then

$$2^{k+1} = 2 \cdot 2^k \geq 2(k+1) \geq k + 2.$$

Strong induction

Sometimes, to prove $P(n)$, we assume

$$P(1), P(2), \ldots, P(n-1)$$

are all true.
This is called **strong induction**.
It is especially useful when the proof of $P(n)$ depends on several smaller cases.

Minimal counterexample method

Another equivalent proof method is the **minimal counterexample method**:

- assume the statement is false,
- choose the smallest counterexample,
- derive a contradiction.

Prime numbers

A **prime number** is an integer $p > 1$ whose only positive divisors are

$$1 \quad \text{and} \quad p.$$

Examples:

$$2, 3, 5, 7, 11, 13, 17.$$

Euclid: **There are infinitely many primes!**

There are infinitely many primes

Suppose there are only finitely many primes: $p_1, p_2, \ldots, p_n$. Set

$$N = p_1 p_2 \cdots p_n + 1.$$

For each $i = 1, \ldots, n$,

$$p_i \nmid N.$$

Thus N is not divisible by any prime in the list.

As $N > 1$, it must have a prime divisor. This gives a prime not among $p_1, \ldots, p_n$, a contradiction.

Composite numbers

An integer $n > 1$ that is not prime is called **composite**.
Examples:

$$9 = 3 \cdot 3, \qquad 12 = 3 \cdot 4, \qquad 15 = 3 \cdot 5.$$

So $9, 12, 15$ are composite.

Two useful remarks

- 2 is the only even prime.
- Every integer $n > 1$ is either prime or composite.

These facts are simple but important.

Prime factorization

To factor an integer means to write it as a product.
A **prime factorization** is a factorization into prime numbers.
Example:

$$30 = 2 \cdot 3 \cdot 5.$$

Example: factor 84

$$84 = 2 \cdot 42$$
$$42 = 2 \cdot 21$$
$$21 = 3 \cdot 7$$

Hence

$$84 = 2^2 \cdot 3 \cdot 7.$$

Another example

$$
\begin{aligned}
90 &= 2 \cdot 45 \\
45 &= 3 \cdot 15 \\
15 &= 3 \cdot 5
\end{aligned}
$$

So

$$90 = 2 \cdot 3^2 \cdot 5.$$

Practice

Find the prime factorizations of

$$72, \qquad 180, \qquad 210.$$

Answers:

$$72 = 2^3 \cdot 3^2, \quad 180 = 2^2 \cdot 3^2 \cdot 5, \quad 210 = 2 \cdot 3 \cdot 5 \cdot 7.$$

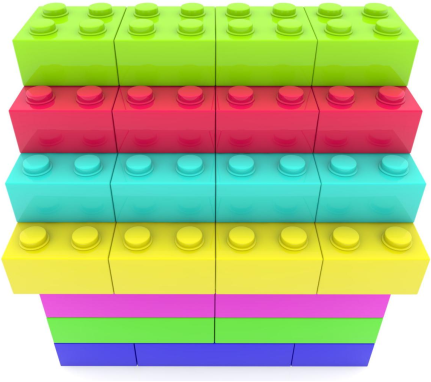
Existence of prime factorization

Theorem. Every integer

$$n > 1$$

can be written as a product of primes.

Prime numbers play a role similar to **atoms** in chemistry: they are the basic components from which all integers are built.



$$60 = 2^2 \cdot 3 \cdot 5$$

Prime numbers are the building blocks of integers.

Proof by strong induction

Base case: for $n = 2$, the statement is true, since 2 is prime.

Assume every integer m with $2 \leq m < n$, can be written as a product of primes.

Now consider n.

- If n is prime, we are done.
- If n is composite, then
$$n = ab$$
  with
$$1 < a < n, \qquad 1 < b < n.$$

Existence proof: finish

By the strong induction hypothesis, both a and b are products of primes.
Therefore

$$n = ab$$

is also a product of primes.
So every integer

$$n > 1$$

has a prime factorization.

Alternative proof idea

The same theorem can also be proved by the minimal counterexample method:

- assume some integer $n > 1$ has no prime factorization,
- choose the smallest such n,
- show that this leads to a contradiction.

Recall: Bézout identity

From last week: for integers a, b, there exist integers x, y such that

$$\gcd(a, b) = ax + by.$$

In particular, if $\gcd(a, b) = 1$, then

$$ax + by = 1.$$

Euclid's Lemma

Lemma. If p is prime and

$$p \mid ab,$$

then

$$p \mid a \quad \text{or} \quad p \mid b.$$

Euclid's Lemma: proof idea

Assume

$$p \mid ab$$

but

$$p \nmid a.$$

Since p is prime, the only positive divisors of p are 1 and p. Thus

$$\gcd(p, a) = 1.$$

Euclid's Lemma: Bézout step

Since

$$\gcd(p, a) = 1,$$

Bézout identity gives integers x, y such that

$$px + ay = 1.$$

Multiply by b:

$$pbx + aby = b.$$

Euclid's Lemma: finish

Now

$$p \mid pbx$$

and

$$p \mid ab,$$

so

$$p \mid aby.$$

Hence p divides the sum

$$pbx + aby = b.$$

Therefore

$$p \mid b.$$

Uniqueness of prime factorization

Theorem. If

$$n > 1$$

has a prime factorization, then that factorization is unique up to order.

Uniqueness proof: setup

Suppose

$$n = p_1 p_2 \cdots p_k$$

and also

$$n = q_1 q_2 \cdots q_m$$

where all p_i and q_j are prime.

Uniqueness proof: first step

Since

$$p_1 \mid n = q_1 q_2 \cdots q_m,$$

Euclid's Lemma implies that p_1 divides one of the primes q_j.
But a prime dividing another prime must be equal to it.
So

$$p_1 = q_j$$

for some j.

Uniqueness proof: finish

Reorder the q_j if necessary and cancel the common prime.
Then repeat the same argument.
Eventually all primes match, so the factorization is unique up to order.

Standard form

It is convenient to write prime factorization in the form

$$n = p_1^{e_1} p_2^{e_2} \cdots p_r^{e_r},$$

where

$$p_1 < p_2 < \cdots < p_r$$

are distinct primes and each $e_i \geq 1$.

Example of standard form

$$360 = 2^3 \cdot 3^2 \cdot 5.$$

Here the distinct primes are

$$2,\ 3,\ 5$$

and the exponents are

$$3,\ 2,\ 1.$$

gcd from prime factorization

If

$$a = p_1^{\alpha_1} \cdots p_r^{\alpha_r}, \qquad b = p_1^{\beta_1} \cdots p_r^{\beta_r},$$

then

$$\gcd(a, b) = p_1^{\min(\alpha_1,\beta_1)} \cdots p_r^{\min(\alpha_r,\beta_r)}.$$

Least common multiple and prime factorizations

The **least common multiple** of two integers a, b, denoted $\mathrm{lcm}(a, b)$, is the smallest positive integer that is divisible by both a and b.

Then

$$\mathrm{lcm}(a, b) = p_1^{\max(\alpha_1, \beta_1)} \cdots p_r^{\max(\alpha_r, \beta_r)}.$$

So gcd uses the smaller exponents, while lcm uses the larger ones.

Example: 36 and 60

$$36 = 2^2 \cdot 3^2, \qquad 60 = 2^2 \cdot 3 \cdot 5.$$

Therefore

$$\gcd(36, 60) = 2^2 \cdot 3 = 12,$$
$$\operatorname{lcm}(36, 60) = 2^2 \cdot 3^2 \cdot 5 = 180.$$

Example: 84 and 126

$$84 = 2^2 \cdot 3 \cdot 7, \qquad 126 = 2 \cdot 3^2 \cdot 7.$$

Thus

$$\gcd(84, 126) = 2 \cdot 3 \cdot 7 = 42,$$

$$\operatorname{lcm}(84, 126) = 2^2 \cdot 3^2 \cdot 7 = 252.$$

Quizz

1. Using prime factorization to compute
$$\gcd(90, 126) \qquad \text{and} \qquad \mathrm{lcm}(90, 126).$$
2. Is there an efficient way to compute $\mathrm{lcm}(a, b)$?

Solution

$$90 = 2 \cdot 3^2 \cdot 5, \qquad 126 = 2 \cdot 3^2 \cdot 7.$$

Hence

$$\gcd(90, 126) = 2 \cdot 3^2 = 18,$$
$$\operatorname{lcm}(90, 126) = 2 \cdot 3^2 \cdot 5 \cdot 7 = 630.$$

Summary

This week we learned:

- induction
- prime numbers and factorization
- Euclid's Lemma
- uniqueness of factorization
- gcd and lcm from prime factorization